

Algebra Exercises

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Abstract

I complete all exercises in Algebra: Chapter 0, and verify against the answer key after the fact.

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Chapter 1

Preliminaries: Set Theory and Categories

1.1 Naive Set Theory

Exercise 1.1.1. Locate a discussion of Russell's paradox, and understand it.

Solution. Russell's paradox shows that a set theory with the Axiom Schema of Unrestricted Comprehension, which says

$$\exists y \forall x (x \in y \iff \varphi(x))$$

for all statements $\varphi(x)$, must be inconsistent. This axiom says the set

$$\{x \mid \varphi(x)\}$$

exists. Russell considered the example $\varphi(x) = x \notin x$. In this case, $x \in x \iff x \notin x$, which is a contradiction.

The axioms of ZFC were formulated in an attempt to get rid of this contradiction. They say only subsets exist; that is, given a set X , any subset $Y \subseteq X$ definable in first-order logic exists. That is, they reformulated set-builder notation such that only sets of the form

$$\{x \in X \mid \varphi(x)\}$$

exist in general. ■

Exercise 1.1.2. Prove that if $\sim$ is an equivalence relation on a set S , then the corresponding family $\mathcal{P}_\sim$ defined as the set of equivalence classes of S under $\sim$ forms a partition of S .

Proof. I swear I've done this so many times now. Let $[a]_\sim, [b]_\sim \in \mathcal{P}_\sim$ such that $[a]_\sim \neq [b]_\sim$. Note that if $a \sim b$, then $a \in [b]_\sim$. By contrapositive, $a \not\sim b$. Suppose for contradiction that $x \in [a]_\sim \cap [b]_\sim$; that is, $[a]_\sim \cap [b]_\sim \neq \emptyset$. It follows that $a \sim x$ and $b \sim x$, and thus $a \sim b \implies \Leftarrow$. Therefore, $[a]_\sim \cap [b]_\sim = \emptyset$. Now let $x \in S$. Note that $x \sim x$, so $x \in [x]_\sim \in \mathcal{P}_\sim$. Therefore, $\mathcal{P}_\sim$ partitions S . ■

Exercise 1.1.3. Given a partition $\mathcal{P}$ on S , show how to define a relation $\sim$ on S such that $\mathcal{P}$ is the corresponding partition.

Solution. Let $\mathcal{P}$ partition S . Define the relation

$$a \sim b \iff \exists X \in \mathcal{P} (a \in X \wedge b \in X).$$

First, we show that $\sim$ is an equivalence relation, as stated. Note that since $\mathcal{P}$ partitions S , there exists some $X \in \mathcal{P}$ such that $a \in X$ for all $a \in S$. Therefore, $a \sim a$ trivially. Now let $a, b \in S$ and suppose $a \sim b$. It follows that there exists $X \in \mathcal{P}$ such that $a, b \in X$. Since $\wedge$ is commutative by truth table, we have $b \sim a$. Now let $a \sim b$ and $b \sim c$. It follows that $a, b \in X_1 \in \mathcal{P}$ and $b, c \in X_2 \in \mathcal{P}$. Since $\mathcal{P}$ is a partition, and $b \in X_1 \cap X_2$, we have $X_1 = X_2$, and thus $a, c \in X_1$. Therefore, $a \sim c$, and $\sim$ is an equivalence relation.

Now let $\mathcal{P}_\sim$ be the partition formed by equivalence classes of S under $\sim$. We will show $\mathcal{P}_\sim = \mathcal{P}$. Let $[a]_\sim \in \mathcal{P}_\sim$, and let $a \in X \in \mathcal{P}$. Note that if $b \in [a]_\sim$, then $a \sim b$, and thus $b \in X$ by definition. Also note that if $b \in X$, then $a \sim b$ by definition, and thus $b \in [a]_\sim$. Therefore, $[a]_\sim = X$, and $[a]_\sim \in \mathcal{P}$. Now let $X \in \mathcal{P}$. It follows that X is nonempty, so let $a \in X$. Note that by the same argument we have $[a]_\sim = X$, so $X \in \mathcal{P}_\sim$. Therefore, $\mathcal{P} = \mathcal{P}_\sim$. ■

Exercise 1.1.4. How many equivalence relations can be defined on the set $\{1, 2, 3\}$?

Solution. As shown previously, the number of equivalence relations is precisely the same as the number of partitions. Note that the only partitions are

$$\{\{1\}, \{2\}, \{3\}\}, \{\{1, 2\}, \{3\}\}, \{\{1\}, \{2, 3\}\}, \{\{1, 3\}, \{2\}\}, \{1, 2, 3\}.$$

Therefore, there are 5 equivalence relations on $\{1, 2, 3\}$. ■

Exercise 1.1.5. Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you attempt to use this relation to define a partition on the set?

Solution. We will answer the second question first. Suppose a relation $\sim$ on a set S is reflexive and symmetric, but not in general transitive. Note that since $\forall x \in S, x \sim x$ still holds, then the classes still cover S . However, they are not necessarily disjoint anymore. For example, let $a \sim b$, $b \sim c$, and $a \not\sim c$. We have $a, c \in [b]_\sim$, but $c \notin [a]_\sim$. Therefore, $[a]_\sim \cap [b]_\sim \neq \emptyset$, but $[a]_\sim \neq [b]_\sim$. Therefore, the classes do not form an equivalence relation in general.

However, note the following. If we have an antitransitive relation, that is $a \sim b$ and $b \sim c$ implies $a \not\sim c$, then we necessarily have $a \not\sim a$ because $a \sim b$ and $b \sim a$. Wow this is tricky. Will come back to it later. ■

Exercise 1.1.6. Define a relation $\sim$ on $\mathbb{R}$ by letting $a \sim b$ if and only if $b - a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a ‘compelling’ description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on $\mathbb{R} \times \mathbb{R}$ defined by $(a_1, a_2) \approx (b_1, b_2)$ if and only if $b_1 - a_1, b_2 - a_2 \in \mathbb{Z}$.

Solution. Note that $\forall x \in \mathbb{R}, x - x = 0 \in \mathbb{Z}$, so $x \sim x$. Suppose $a \sim b$. Then $b - a \in \mathbb{Z}$. Note that $a - b = -(b - a)$, and thus $b - a \in \mathbb{Z}$. Therefore, $a \sim b \implies b \sim a$. Now suppose $a \sim b$ and $b \sim c$. Since $b - a \in \mathbb{Z}$, we have $b - a \in \mathbb{Z}$ and $c - b \in \mathbb{Z}$. Note that $\mathbb{Z}$ is closed under addition, so we have

$$(c - b) + (b - a) = c - a \in \mathbb{Z}.$$

Therefore, $c - a \in \mathbb{Z}$, and $\sim$ is an equivalence relation.

The set $\mathbb{R}/\sim$ can be thought of as the interval $[0, 1)$. Note that we can think of any real number x as having an integer component, $\lfloor x \rfloor$, along with a non-integer component $x - \lfloor x \rfloor$. By definition, we have $0 \leq x - \lfloor x \rfloor < 1$, and also note that $a \sim b$ if and only if they have an

equivalent non-integer component. Therefore, we can think of this set by chopping off everything in front of the decimal place, and only considering the non-integer component.

I will breeze through $\approx$ since it's basically the same. $(a, b) - (a, b) = (0, 0) \in \mathbb{Z}^2$. $-((a_1, b_1) - (a_2, b_2)) = (a_2, b_2) - (a_1, b_1) \in \mathbb{Z}^2$. I'm not even going to type the third one out it's literally the same. The set $\mathbb{R}^2/\sim$ is the set of all pairs where the non-integer component is the same for both left components, and both right components.

NOTE: for negative numbers it should be $\lceil \cdot \rceil$, not $\lfloor \cdot \rfloor$. ■

1.2 Functions

Exercise 1.2.1. How many different bijections are there from a set S with n elements to itself?

Solution. This is the number of permutations of a set, which is well known to be $n!$, since a permutation of a finite set corresponds directly to an ordering of the elements of the set. ■

Exercise 1.2.2. Prove statement (2) in Proposition 2.1 (1.1). That is, prove that f has a right-inverse if and only if it is surjective.

Proof. Let $f : A \rightarrow B$. Suppose there exists $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$. For all $b \in B$, we have

$$b = \text{id}_B(b) = (f \circ g)(b) = f(g(b)).$$

Since $g(b) \in A$, then for all $b \in B$ there exists $a := g(b)$ in A such that $f(a) = b$, and f is surjective.

Now we show the converse. Let $f : A \rightarrow B$ be surjective. Then for all $b \in B$, there exists $a \in A$ such that $f(a) = b$. Define $\mathcal{P}_f$ to be the set of equivalence classes in B formed by f . By the Axiom of Choice, there exists a set $\mathcal{A}$ consisting of precisely one chosen element $a \in [a]_f$, for all $[a]_f \in \mathcal{P}_f$. Note that $\mathcal{A}$ can be viewed as a subset of A via an inclusion map, so without loss of generality consider $\mathcal{A} \subseteq A$. Define $g : B \rightarrow A$ as the map

$$g(b) = a \iff a \in \mathcal{A} \wedge f(a) = b.$$

By the Axiom of Choice and since f is surjective, g is well defined. By definition of g , we have

$$(f \circ g)(b) = f(g(b)) = f(a) = b.$$

Therefore, f has a right-inverse if and only if f is surjective. ■

Exercise 1.2.3. Prove that the inverse of a bijection is a bijection and that the composition of two bijections is a bijection.

Proof. Let $f : A \rightarrow B$ be a bijection. Since f is a bijection, it has an inverse $f^{-1} : B \rightarrow A$. Note that $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$, so $(f^{-1})^{-1} = f$ is a two-sided inverse. Therefore, f^{-1} is a bijection.

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections. Therefore, there exist $f^{-1} : B \rightarrow A$ and $g^{-1} : C \rightarrow B$ such that

$$f^{-1} \circ f = \text{id}_A, \quad f \circ f^{-1} = \text{id}_B, \quad g^{-1} \circ g = \text{id}_B, \quad g \circ g^{-1} = \text{id}_C.$$

Consider the map $f^{-1} \circ g^{-1}$. We wish to show $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. Note that

$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = (g \circ \text{id}_B) \circ g^{-1} = g \circ g^{-1} = \text{id}_C.$$

Additionally,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ (\text{id}_B \circ f) = f^{-1} \circ f = \text{id}_A.$$

Therefore, $g \circ f$ has $f^{-1} \circ g^{-1}$ as a two-sided inverse, and therefore $g \circ f$ is a bijection. ■

Exercise 1.2.4. Prove that isomorphism is an equivalence relation on any set of sets.

Proof. Let S be a set of sets, and let $\cong$ be the relation such that $A \cong B$ if and only if there exists a bijection $A \xrightarrow{\sim} B$.

Note that for all $A \in S$, $\text{id}_A : A \rightarrow A$ is a bijection, so $A \cong A$. Let $A, B \in S$ such that $A \cong B$. By definition, there exists a bijection $f : A \rightarrow B$. It follows from proposition 1.1 that there exists $f^{-1} : B \rightarrow A$ such that $f \circ f^{-1} = \text{id}_B$ and $f^{-1} \circ f = \text{id}_A$. By exercise 1.2.3, f^{-1} is a bijection. Therefore, $B \cong A$. Finally, let $A, B, C \in S$ such that $A \cong B$ and $B \cong C$. By definition, there exist bijections $f : A \rightarrow B$ and $g : B \rightarrow C$. By exercise 1.2.3, there exists a bijection $g \circ f : A \rightarrow C$, so $A \cong C$. Therefore, $\cong$ is an equivalence relation on S . ■

Exercise 1.2.5. Formulate the notion of an *epimorphism*, in the style of an *monomorphism* as seen in section 1.2.6, and prove an analogous result to proposition 2.3 (1.2), for epimorphisms and surjections.

Proof. Let $f : A \rightarrow B$ be a function. Then f is an *epimorphism* if and only if for all sets X and all functions $\beta', \beta'' : B \rightarrow X$,

$$\beta' \circ f = \beta'' \circ f \implies \beta' = \beta''.$$

We will now prove a function is epic if and only if it is surjective. Let f be surjective. By exercise 1.2.2, there exists a map $g : B \rightarrow A$ such that $g \circ f = \text{id}_B$. It follows that

$$\begin{aligned} \beta' \circ f = \beta'' \circ f &\implies (\beta' \circ f) \circ g = (\beta'' \circ f) \circ g \\ &\implies \beta' \circ (f \circ g) = \beta'' \circ (f \circ g) \\ &\implies \beta' \circ \text{id}_B = \beta'' \circ \text{id}_B \\ &\implies \beta' = \beta''. \end{aligned}$$

Therefore, f is an epimorphism. Now we show the converse. Let f be an epimorphism. Suppose for contradiction that f is not surjective. It follows that there exists at least one $b \in B$ such that for all $a \in A$, $f(a) \neq b$. Since f is an epimorphism, for all sets X , there exist $\beta', \beta'' : B \rightarrow X$ such that

$$\beta' \circ f = \beta'' \circ f \implies \beta' = \beta''.$$

Let $x = \{0, 1\}$, and let

$$\beta'(x) := 1, \quad \beta'' := \begin{cases} 0, & x = b \\ 1, & x \neq b \end{cases}.$$

Note that $\beta' \neq \beta''$. However, since $b \notin f(A)$, we have

$$\forall a \in A : \beta'(f(a)) = \beta''(f(a)).$$

Therefore, $\beta' \neq \beta''$, which is a contradiction. Therefore, f must be surjective, and thus a map is epic if and only if it is surjective. ■

Exercise 1.2.6. With notation as previously, explain how any function $f : A \rightarrow B$ determines a section of π_A .

Solution. Note that a *section* is simply a right inverse of a surjection. Let $\pi_A : A \times B \rightarrow A$ be the natural projection $(a, b) \mapsto a$, and let $f : A \rightarrow B$ be any function. A right inverse ϕ_f of π is a map $\phi_f : A \rightarrow A \times B$ such that $\pi \circ \phi_f = \text{id}_A$. Define $\phi_f a = (a, f(a))$. For all $a \in$, we have

$$(\pi \circ \phi_f)(a) = \pi(\phi_f(a)) = \pi(a, f(a)) = a.$$

Therefore, $\pi \circ \phi = \text{id}_A$. Therefore, every function $f : A \rightarrow B$ determines a section ϕ_f of π_A in this manner. ■

Exercise 1.2.7. Let $f : A \rightarrow B$ be any function. Show that f is isomorphic to A .

Proof. Define $\pi_A : f \rightarrow A$ to be the map $(a, b) \mapsto a$. We will show π_A is a bijection. Since f is a function, then for all $a \in A$ there exists a pair $(a, b) \in f$. Note that $\pi_A(a, b) = a$. Therefore, π_A is surjective. Now let $\pi_A(a_1, b_1) = \pi_A(a_2, b_2)$. It follows that $a_1 = a_2$, and by definition of a function, for each $a \in A$ there exists exactly one $b \in B$ such that $(a, b) \in f$. Therefore, $b_1 = b_2$, and thus $(a_1, b_1) = (a_2, b_2)$. Therefore, π is injective, and thus an isomorphism. Therefore, $f \cong A$. ■

Exercise 1.2.8. Describe as explicitly as you can all terms in the canonical decomposition of the function $\mathbb{R} \rightarrow \mathbb{C}$ defined by $r \mapsto e^{2\pi i r}$.

Solution. Note that if $\sim$ is the equivalence relation determined on $\mathbb{R}$ by f , then by our knowledge of $\mathbb{C}$ we can say $x \sim y$ if and only if $|x - [x]| = |y - [y]|$. We can then represent each equivalence class with a real number $x \in [0, 1)$. We then have the canonical projection $\mathbb{R} \twoheadrightarrow \mathbb{R}/\sim$, which can be defined as $x \mapsto [x - [x]]_\sim$. We can then define a bijection $\mathbb{R}/\sim \xrightarrow{\sim} \mathbb{T}$ via $[x]_\sim \mapsto e^{2\pi i x}$, and then the inclusion map $\iota : \mathbb{T} \rightarrow \mathbb{C}$ defined by $z \mapsto z$ is an injection. ■

Exercise 1.2.9. Show that if $A' \cong A''$ and $B' \cong B''$, and further $A' \cap B' = \emptyset$ and $A'' \cap B'' = \emptyset$, then $A' \cup B' \cong A'' \cup B''$. Conclude that the operation $A \amalg B$ is well-defined up to isomorphism.

Proof. Suppose $A' \cong A''$, $B' \cong B''$, $A' \cap B' = \emptyset$, and $A'' \cap B'' = \emptyset$. We will define a bijection $A' \cup B' \rightarrow A'' \cup B''$. Since $A' \cong A''$, there exists a bijection $f_A : A' \rightarrow A''$. Similarly, there exists a bijection $f_B : B' \rightarrow B''$. Define $g : A' \cup B' \rightarrow A'' \cup B''$ as the map

$$g(x) = \begin{cases} f_A(x), & x \in A' \\ f_B(x), & x \in B' \end{cases}.$$

Since $A' \cap B' = \emptyset$, the function is well defined. First, we show surjectivity. Let $x \in A'' \cup B''$. Since $A'' \cap B'' = \emptyset$, we have either $x \in A''$ or $x \in B''$, but never $x \in A'' \cap B''$. In either case, since f_A, f_B are bijections, and thus surjections, there exists $x' \in A' \cup B'$ such that $g(x') = x$. Injectivity also follows trivially from the fact that A'' and B'' are disjoint and isomorphic to A' and B' . Therefore, $A'' \cup B'' \cong A' \cup B'$.

Note that if A', B' and A'', B'' are disjoint isomorphic copies of A and B , then any choice of isomorphic copies for the operation $A \amalg B$ is isomorphic to any other choice. Therefore, we say that the operation is well-defined up to isomorphism, since all choices are equivalent up to

| isomorphism. ■

Exercise 1.2.10. Show that if A and B are finite sets, then the set of functions $A \rightarrow B$, denoted B^A , has $|B^A| = |B|^{|A|}$.

| **Proof.** I am not good at counting bro ■

Exercise 1.2.11. In view of exercise 1.2.10, it's not unreasonable to use 2^A to denote the set of functions from an arbitrary set A to a set with two elements $\{0, 1\}$. Prove that there exists a bijection between 2^A and $\mathcal{P}(A)$.

| **Proof.** I want to remark that in ZFC, it's more than perfectly reasonable to say this given the previous notation, since the Von-Neumann ordinal 2 is defined as $\{0, 1\}$. Anyways, here is the bijection. For all $S \in \mathcal{P}(A)$, define $f_S : A \rightarrow 2$ as the map

$$f_S(x) = \begin{cases} 1, & x \in S \\ 0, & x \notin S \end{cases}.$$

Clearly, if this map is well-defined, then it is surjective. We must first show that all functions $f \in 2^A$ can be represented in this manner. Note that for all $f \in 2^A$, there is an induced equivalence relation $\sim$ on A such that $a \sim b$ if and only if $f(a) = f(b)$. There are precisely two of these equivalence classes, corresponding to 0 and 1. Denote these as $[0]$ and $[1]$. Note that $[1]$, even if it is empty, is a subset of A , and thus $[1] \in \mathcal{P}(A)$. Therefore, $f = f_{[1]}$, and thus all functions can be represented as some f_S for $S \in \mathcal{P}(A)$.

Define $\phi : 2^A \rightarrow \mathcal{P}(A)$ as the map $f_S \mapsto S$. This map is again, clearly surjective, so we show only that it is injective. Let $f_S \neq f_T$. Since there are only two elements to map to, a change in the set of elements mapping to 0 is equivalent to a change in the set of elements mapping to 1, and thus there must be a change in the set of elements mapping to 1. Therefore, $S \neq T$, and ϕ is bijective.

A much easier proof, without identifying the precise bijection, is to simply say that since $|\mathcal{P}(A)| = 2^{|A|}$, we have $|2^A| = |\mathcal{P}(A)|$, and thus a bijection exists by exercise 1.2.10. ■

1.3 Categories

Exercise 1.3.1. Let $\mathbf{C}$ be a category. Consider a structure $\mathbf{C}^{op}$ with

- $\text{Obj}(\mathbf{C}^{op}) := \text{Obj}(\mathbf{C})$;
- for A, B objects of $\mathbf{C}^{op}$, $\text{Hom}_{\mathbf{C}^{op}}(A, B) := \text{Hom}_{\mathbf{C}}(B, A)$.

Show how to make this into a category (that is, define composition of morphisms in $\mathbf{C}^{op}$ and verify the properties in the definition).

| **Solution.** If $f : A \rightarrow B$ and $g : B \rightarrow C$ in $\mathbf{C}^{op}$, then $f : B \rightarrow A$ and $g : C \rightarrow B$ in $\mathbf{C}$. This implies $fg : C \rightarrow A$ in $\mathbf{C}$, so we can define gf in $\mathbf{C}^{op}$ to be fg in $\mathbf{C}$. We will show now that this definition, together with the above definitions, satisfies the properties of a category.

Notice that $\text{End}_{\mathbf{C}}(A) = \text{End}_{\mathbf{C}^{op}}(A)$, so there exists $1_A \in \text{End}_{\mathbf{C}^{op}}(A)$ trivially. Let $f \in \text{Hom}_{\mathbf{C}^{op}}(A, B)$ and $g \in \text{Hom}_{\mathbf{C}^{op}}(B, C)$. Note that $f \in \text{Hom}_{\mathbf{C}}(B, A)$ and $g \in \text{Hom}_{\mathbf{C}}(C, B)$, and thus $fg \in \text{Hom}_{\mathbf{C}}(C, A)$. Therefore, there exists $h \in \text{Hom}_{\mathbf{C}^{op}}(A, C)$ such that $h = fg$. By our previous definition, $h = gf$. Therefore, $gf \in \text{Hom}_{\mathbf{C}^{op}}(A, C)$.

Let A, B, C, D be objects in $\mathbf{C}^{op}$. Let $f \in \text{Hom}_{\mathbf{C}^{op}}(A, B)$, $g \in \text{Hom}_{\mathbf{C}^{op}}(B, C)$, and $h \in \text{Hom}_{\mathbf{C}^{op}}(C, D)$. Let f', g', h' be the corresponding functions to f, g, h in $\mathbf{C}$. Note that $gf = f'g'$ and $hg = g'h'$. We have $h(gf) = (f'g')h' = f'(g'h') = (hg)f$. Therefore, composition is associative. Finally, that $\mathbb{1}_A$ is an identity follows trivially from $\mathbf{C}$ being a category. ■

Exercise 1.3.2. If A is a finite set, how large is $\text{End}_{\mathbf{Set}}(A)$?

Solution. This is basically asking how many functions $f : A \rightarrow A$ exist. Since the set is finite, we can do the following. First, fix some number $n \leq |A|$, and choose some subset $S \subseteq A$ of cardinality n . Next, find all the distinct ways to partition A into n nonempty subsets. Assign each subset in the partition to an element of the range. There are $n!$ ways to do this, so we multiply through by $n!$. Now find all ways to choose a range with n elements, which is $C(|A|, n)$. Now multiply all of these together and sum over all $1 \leq n \leq |A|$. I believe this is

$$\sum_{n=1}^{|A|} n! \cdot S(|A|, n) \cdot \binom{|A|}{n}.$$

Exercise 1.3.3. Formulate precisely what it means to say that $\mathbb{1}_a$ is an identity with respect to composition, and prove this assertion, in the construction of a category as a relation $\sim$ on a set S .

Solution. Let $\hat{\mathbf{S}}$ be a category with $\text{Obj}(\hat{\mathbf{S}}) = S$ for some set S , and $\text{Hom}_{\hat{\mathbf{S}}}(a, b)$ defined as $\{(a, b)\}$ iff $a \sim b$, and $\emptyset$ otherwise. Note that we defined function composition such that if $f = (a, b)$ and $g = (b, c)$, then $gf = (a, c)$. We then have the function composition rule $(b, c) \circ (a, b) := (a, c)$. Now consider $f = (a, b)$ and $\mathbb{1}_a = (a, a)$. We then have $f\mathbb{1}_a = (a, b) \circ (a, a) = (a, b)$. Additionally, if we have $\mathbb{1}_b = (b, b)$, then $\mathbb{1}_b f = (b, b) \circ (a, b) = (a, b)$. I think it's apparent that this means if $a \sim b$ and $b \sim b$ then the transitive property $a \sim b$ implies the same statement. ■

Exercise 1.3.4. Can we define a category of the same style using the relation $<$ on the set $\mathbb{Z}$?

Solution. We cannot. Since $a \not< a$, we would have $\text{End}(A) = \emptyset$, which is not consistent with the definition of a category. ■

Exercise 1.3.5. Explain in what instance the category “ $\text{Obj}(\hat{\mathbf{S}}) := \mathcal{P}(S)$ for S a set, and $\text{Hom}_{\hat{\mathbf{S}}}(A, B) = \{(A, B)\}$ if $A \subseteq B$, and $\emptyset$ otherwise” is an instance of the categories explored as a set S with a relation $\sim$.

Solution. Subset inclusion is in fact a relation, and it has both the reflexive and transitive properties, but is not symmetric. That is, $A = B$ if and only if $A \sim B$ and $B \sim A$. This is all the example required, so we are done. ■

Exercise 1.3.6. Define a category $\mathbf{V}$ by taking $\text{Obj}(\mathbf{V}) := \mathbb{N}$ and letting $\text{Hom}_{\mathbf{V}}(n, m)$ be the set of $m \times n$ matrices with real entries, for all $m, n \in \mathbb{N}$ (including 0). Use product of matrices to define composition. Does this category feel familiar?

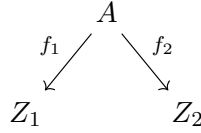
Solution. Recall that an $n \times m$ matrix times an $m \times p$ matrix yields an $n \times p$ matrix. It only makes sense to define a single morphism $0 \rightarrow 0$, which can be given by the empty matrix I guess.

Ok this makes sense. n represents the vector space $\mathbb{R}^n$, and the morphisms are precisely the linear transformations between $\mathbb{R}$ -spaces. That is, a morphism $n \rightarrow m$ corresponds to a linear

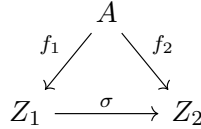
transformation $\mathbb{R}^m \rightarrow \mathbb{R}^n$. In this case, morphisms $n \rightarrow 0$ correspond to the 0 matrix of dimension $1 \times n$, which maps $\mathbb{R}^n \rightarrow \{0\}$, the space with dimension 0. ■

Exercise 1.3.7. Define carefully objects and morphisms in $\mathbf{C}^A$, and draw the diagram corresponding to the composition.

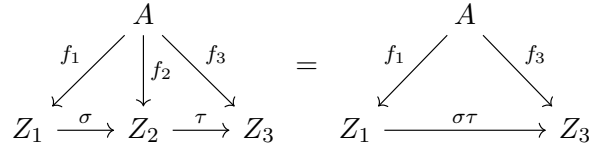
Solution. The objects of $\mathbf{C}^A$ are morphisms $f : A \rightarrow Z$ for any arbitrary $Z \in \text{Obj}(\mathbf{C})$. For example, two objects $f_1, f_2 \in \text{Obj}(\mathbf{C}^A)$ can be represented with the diagram



A morphism $f_1 \rightarrow f_2$ would be a morphism $\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2)$ such that $f_2 = \sigma f_1$. That is,



Composition of morphisms $f_1 \rightarrow f_2 \rightarrow f_3$ looks like the diagram



Exercise 1.3.8. A *subcategory* $\mathbf{C}'$ of a category $\mathbf{C}$ consists of objects of $\mathbf{C}$ with sets of morphisms $\text{Hom}_{\mathbf{C}'}(A, B) \subseteq \text{Hom}_{\mathbf{C}}(A, B)$ for all $A, B \in \text{Obj}(\mathbf{C}')$, such that identities and compositions in $\mathbf{C}$ make $\mathbf{C}'$ into a category. A subcategory is *full* if $\text{Hom}_{\mathbf{C}'}(A, B) = \text{Hom}_{\mathbf{C}}(A, B)$ for all $A, B \in \text{Obj}(\mathbf{C}')$. Construct a category of *infinite sets*, and explain how it may be viewed as a full subcategory of **Set**.

Solution. Let $\mathbf{C}'$ be the structure with $\text{Obj}(\mathbf{C}')$ containing all infinite sets, $\text{Hom}_{\mathbf{C}'}(A, B) = \text{Hom}_{\mathbf{Set}}(A, B)$ for all $A, B \in \text{Obj}(\mathbf{C}')$. Fairly obviously, $\mathbf{C}'$ is a category. It inherits all properties of morphisms from **Set**, and since we only consider morphisms between infinite sets here, composition holds. Note that since $\text{Obj}(\mathbf{C}')$ is contained within $\text{Obj}(\mathbf{Set})$, and since our morphisms are precisely those in **Set**, $\mathbf{C}'$ forms a full subcategory of **Set**. ■

Exercise 1.3.9. An alternative definition to the notion of *multiset* introduced in 2.2 (1.2.2) is obtained by considering sets endowed with equivalence relations; equivalent elements are taken to be multiple instances of elements ‘of the same kind’. Define a notion of morphism between such enhanced sets, obtaining a category **MSet** containing a copy of **Set** as a full subcategory. Which objects in **MSet** determine ordinary multisets as seen in 2.2 and how? Spell out what a morphism of multisets would be from this point of view. Try to define morphisms in **MSet** so that the notion you obtain for ordinary multisets captures your intuitive understanding of these objects.

Solution. One possible definition is here. **MSet** is the set of epimorphisms in **Set**. A multiset can be encoded purely in the information given by a surjection of sets. If A, B are sets and $f : A \twoheadrightarrow B$ is a surjection, then the equivalence relation induced on A by f can be used as an indexing set, and the element each set is mapped to is used as the element to be multiplied. If an equivalence class under f $[a]_{\sim} \subseteq A$ all map to $b \in B$ under f , then we say b appears in the multiset once per element of S ; that is, $|S|$ times. For example, consider the map $\{1, 2, 3\} \twoheadrightarrow \{1, 2\}$ defined by $1, 2 \mapsto 1$ and $3 \mapsto 2$. Our equivalence classes are $\{1, 2\}$ and $\{3\}$. Since $\{1, 2\}$ has cardinality 2 and corresponds to 1, we will see two 1s. Similarly, we will see one 2. So our multiset represented by this function is $\{1, 1, 2\}$. Ordinary multisets as seen earlier are then represented by surjections $A \twoheadrightarrow B \subseteq \mathbb{N}$, although there are now many equivalent copies of each multiset.

Sets are denoted by bijections. Under a bijection $A \rightarrow B$, each equivalence class is a singleton, so each element of B appears precisely once in the multiset. In this way, **Set** can be seen as a subcategory of **MSet**.

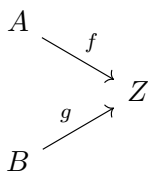
I believe a morphism between multisets should idk ill come back to it after eating ■

Exercise 1.3.10. Since the objects of a category $\mathbf{C}$ are not necessarily sets or similar, it's not generally clear how to make sense of the notion of a subobject. However, in some situations it does make sense to consider subobjects, and the subobjects of any given object A in $\mathbf{C}$ are in one-to-one correspondence with the morphisms $A \rightarrow \Omega$ for a special, fixed object Ω in $\mathbf{C}$, called a *subobject classifier*. Show **Set** has a subobject classifier.

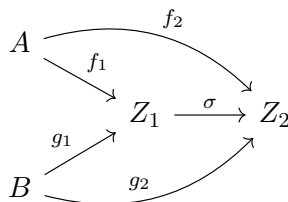
Proof. I have no idea how to do this. Recall that $|B^A| = |B|^{|A|}$. We require $|B|^{|A|} = |A|$, and it follows that $|B| = |A| e^{|A|}$ which is not an integer for any $A \neq \emptyset$. The conclusion is I am misunderstanding what a subobject classifier is. If it's a typo and should be $\Omega \rightarrow A$, then it's trivially any singleton set. ■

Exercise 1.3.11. Draw the relevant diagrams and define composition and identities for the category $\mathbf{C}^{A,B}$ mentioned in example 3.9. Do the same for the category $\mathbf{C}^{\alpha,\beta}$ mentioned in example 3.10.

Solution. We will begin with $\mathbf{C}^{A,B}$. Objects in $\mathbf{C}^{A,B}$ look like pairs of morphisms

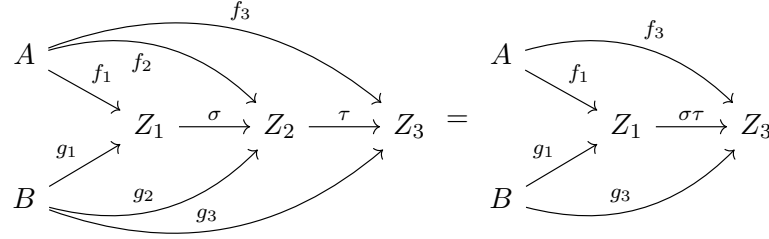


morphisms $(f_1, g_1) \rightarrow (f_2, g_2)$ look like commutative diagrams

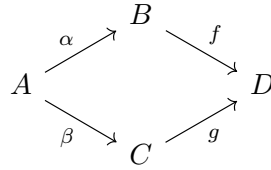


That is, morphisms in $\mathbf{C}^{A,B}$ correspond to morphisms $\sigma \in \text{Hom}_{\mathbf{C}}(Z_1, Z_2)$ such that $f_2 = \sigma f_1$ and

$g_2 = \sigma g_1$. Composition of morphisms $(f_1, g_1) \rightarrow (f_2, g_2) \rightarrow (f_3, g_3)$ looks like diagrams



Now we will do the fibered version. Let $\alpha, \beta : A \rightarrow B, C$ be morphisms in $\mathbf{C}$. Objects in the category $\mathbf{C}^{\alpha, \beta}$ look, once again, like pairs of morphisms $f, g : B, C \rightarrow D$ such that the diagram



commutes. Morphisms $(f_1, g_1) \rightarrow (f_2, g_2)$ look like commutative diagrams

■

1.4 Morphisms

Exercise 1.4.1. Composition is defined for *two* morphisms. If more than two morphisms are given, e.g.,

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{i} E ,$$

then one may compose them in several ways, for example:

$$(ih)(gf), \quad (i(hg))f, \quad i((hg)f), \quad \text{etc.}$$

Prove that the result of any such nested composition is independent of the placement of the parentheses.

Exercise 1.4.2. Let $A, B \in \text{Obj}(\mathbf{C})$, and let $f \in \text{Hom}_{\mathbf{C}}(A, B)$ be a morphism.

- Prove that if f has a right-inverse, then f is an epimorphism.
- Prove that the converse does not hold, by giving an explicit example of a category and an epimorphism without a right-inverse.

Proof. Suppose there exists $g \in \text{Hom}_{\mathbf{C}}(B, A)$ such that $fg = \mathbb{1}_B$. Let $\beta', \beta'' : B \rightarrow X$ for $X \in \text{Obj}(\mathbf{C})$ arbitrary. Suppose $\beta' f = \beta'' f$. Then

$$(\beta' f)g = (\beta'' f)g \implies \beta'(fg) = \beta''(fg) \implies \beta' \mathbb{1}_B = \beta'' \mathbb{1}_B \implies \beta' = \beta''.$$

Therefore, f is an epimorphism.

■

Exercise 1.4.3. Prove that the composition of two monomorphisms is a monomorphism. Deduce that one can define a subcategory $\mathbf{C}_{\text{mono}}$ of a category $\mathbf{C}$ to be the subset of $\text{Hom}_{\mathbf{C}}(A, B)$ consisting of monomorphisms, for all $A, B \in \text{Obj}(\mathbf{C})$. Do the same for epimorphisms. Can you define a subcategory $\mathbf{C}_{\text{nonmono}}$ of $\mathbf{C}$ by restricting morphisms that are *not* monomorphisms?

Exercise 1.4.4. Give a concrete description of monomorphisms and epimorphisms in the category $\mathbf{MSet}$.

1.5 Universal Properties

Exercise 1.5.1. Prove that a final object in a category $\mathbf{C}$ is initial in the opposite category $\mathbf{C}^{op}$.

Proof. Let F be final in $\mathbf{C}$. It follows that $|\mathrm{Hom}_{\mathbf{C}}(-, F)| = 1$ for all objects in $\mathbf{C}$. By definition of $\mathbf{C}^{op}$, $\mathrm{Hom}_{\mathbf{C}^{op}}(F, -) = \mathrm{Hom}_{\mathbf{C}}(-, F)$. Therefore, $|\mathrm{Hom}_{\mathbf{C}^{op}}(F, -)| = 1$ for all objects in $\mathbf{C}^{op}$, and thus F is initial in $\mathbf{C}^{op}$.

Interesting aside: $(\mathbf{C}^{op})^{op} = \mathbf{C}$ pretty trivially, so this shows all initial objects in $\mathbf{C}$ are final in $\mathbf{C}^{op}$ without having to repeat the proof. ■

Exercise 1.5.2. Prove that $\emptyset$ is the *unique* initial object in $\mathbf{Set}$.

Proof. From our knowledge of sets, we know the only set with $|S| = 0$ is $S = \emptyset$. Therefore, $[\emptyset]_{\cong} = \{\emptyset\}$, so it suffices to show $\emptyset$ is initial. Since the only function that can be defined $\emptyset \rightarrow A$ is the empty function, and since there exists an empty function $\emptyset \rightarrow A$ for all $A \in \mathrm{Obj}(\mathbf{Set})$, $\emptyset$ is initial in $\mathbf{Set}$. ■

Exercise 1.5.3. Prove that final objects are unique up to isomorphism.

Proof. Let F_1, F_2 be final in $\mathbf{C}$. Then there exists precisely one morphism $f_1 : F_1 \rightarrow F_2$, and precisely one morphism $f_2 : F_2 \rightarrow F_1$. Since $f_2 f_1 \in \mathrm{End}_{\mathbf{C}}(F_1) = \{\mathbb{1}_{F_1}\}$, we have $f_2 f_1 = \mathbb{1}_{F_1}$. Applying the same logic in the other direction shows $f_1^{-1} = f_2$, and thus f_1, f_2 are isomorphisms, and $F_1 \cong F_2$. ■

Exercise 1.5.4. What are initial and final objects in the category of pointed sets $\mathbf{Set}^*$? Are they unique?

Solution. Singletons $Q = \{q\}$ constitute both the initial and final objects of $\mathbf{Set}^*$. Recall that objects in this category are really morphisms $\{*\} \rightarrow S$ for S a set, but we represent them with pairs (S, s) to state that $* \mapsto s$. Morphisms $(S, s) \rightarrow (T, t)$ in this category are maps $S \rightarrow T$ such that $s \mapsto t$. Note that for all singletons Q , there is precisely one map $S \rightarrow Q$, the constant map. Therefore, singletons constitute the final objects. Moreover, for all singletons (Q, q) , there is precisely one map from $Q \rightarrow S$ such that $q \mapsto s$ for each specific $s \in S$, so there is one morphism $(Q, q) \rightarrow (S, s)$. Therefore, singletons are both initial and final. ■

Exercise 1.5.5. What are initial and final objects in the category $\mathbf{C}^A$ for A a set?

Exercise 1.5.6. Consider the category corresponding to endowing (as in example 3.3) the set $\mathbb{Z}^+$ of positive integers with the *divisibility* relation. Thus there is only one morphism $d \rightarrow m$ in this category if and only if $d|m$; there is no morphism otherwise. Show that this category has products and coproducts. What are their “conventional” names?

Exercise 1.5.7. Redo exercise 1.2.9 using proposition 5.4 (1.6).

Exercise 1.5.8. Show that in every category $\mathbf{C}$ the products $A \times B$ and $B \times A$ are isomorphic, if they exist. (Hint: Observe that they both satisfy the universal property for the product of A and B , then use proposition 5.4 (1.6)).

Exercise 1.5.9. Let $\mathbf{C}$ be a category with products. Find a reasonable candidate for the universal property that the product $A \times B \times C$ of *three* objects of $\mathbf{C}$ ought to satisfy, and prove both $(A \times B) \times C$ and $A \times (B \times C)$ satisfy this property. Deduce that $(A \times B) \times C$ and $A \times (B \times C)$ are isomorphic.

Exercise 1.5.10. Define products and coproducts for *families* (i.e., indexed sets) of objects of a category. Do these exist in **Set**? It is common to denote the product $A \times \cdots \times A$ by A^n .

Exercise 1.5.11. Let A, B be sets, endowed with equivalence relations $\sim_A, \sim_B$, respectively. Define a relation $\sim$ on $A \times B$ by setting

$$(a_1, b_1) \sim (a_2, b_2) \iff a_1 \sim_A a_2 \wedge b_1 \sim_B b_2.$$

- Use the universal property for quotients as seen earlier to establish that there are functions $(A \times B)/\sim \rightarrow A/\sim_A$, $(A \times B)/\sim \rightarrow B/\sim_B$.
- Prove that $(A \times B)/\sim$, with these two functions, satisfies the universal property for the product of $A/\sim_A$ and $B/\sim_B$.
- Conclude (without further work) that $(A \times B)/\sim \cong (A/\sim_A) \times (B/\sim_B)$.

Exercise 1.5.12. Define the notions of *fibered products* and *fibered coproducts*, as terminal objects of the categories $\mathbf{C}_{\alpha, \beta}$ and $\mathbf{C}^{\alpha, \beta}$ by stating carefully the corresponding universal properties. As it happens, **Set** has both fibered products and coproducts. Define these objects concretely, in terms of naive set theory.